

Log-Linear Model (로그선형 모형)

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Today's Topics

Log-Linear Model

- Log-Linear Model for Contingency Table
- Log-Linear Model of Three-Way Contingency Table
- Goodness of fit test
- Relationship between Log Linear Model and Logistic Model

Log-Linear Model for Contingency Table

Log-linear model

Property of log-linear model for two-way contingency table

It models association between categorical variables and their interaction for contingency table

Poisson distribution with log link function is considered

There are no response variables, and all variables are considered as explanatory variables

Reparameterization

	Y				
	levels	1	...	J	sum
X	1	$\mu_{11} = e^{\lambda + \lambda_1^X + \lambda_1^Y + \lambda_{11}^{XY}}$	...	$\mu_{1J} = e^{\lambda + \lambda_1^X + \lambda_J^Y + \lambda_{1J}^{XY}}$	μ_{1+}
	$\vdots$	$\vdots$		$\vdots$	$\vdots$
	I	$\mu_{I1} = e^{\lambda + \lambda_I^X + \lambda_1^Y + \lambda_{I1}^{XY}}$	...	$\mu_{IJ} = e^{\lambda + \lambda_I^X + \lambda_J^Y + \lambda_{IJ}^{XY}}$	μ_{I+}
	sum	μ_{+1}	...	μ_{+J}	n

Log-Linear Model for Contingency Table (Cont'd)

Estimates

Log-linear model: $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$

(1) Sum-to-zero restriction: $\sum_{i=1}^I \lambda_i^X = \sum_{j=1}^J \lambda_j^Y = \sum_{i=1}^I \lambda_{ij}^{XY} = \sum_{j=1}^J \lambda_{ij}^{XY} = 0$

(2) 0-1 restriction: $\lambda_I^X = \lambda_J^Y = \lambda_{iJ}^{XY} = \lambda_{Ij}^{XY} = 0$

Parameter estimators under sum-to-zero restriction

$$\eta_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}, \quad \eta_{i.} = \frac{1}{J} \sum_{j=1}^J \eta_{ij}, \quad \eta_{.j} = \frac{1}{I} \sum_{i=1}^I \eta_{ij}, \quad \mu = \eta_{..} = \frac{1}{IJ} \sum_{j=1}^J \sum_{i=1}^I \eta_{ij}$$

$$\Rightarrow \lambda = \eta_{..}, \quad \lambda_i^X = \eta_{i.} - \eta_{..}, \quad \lambda_j^Y = \eta_{.j} - \eta_{..}, \quad \lambda_{ij}^{XY} = \eta_{ij} - \eta_{i.} - \eta_{.j} + \eta_{..}$$

Log-Linear Model for Contingency Table (Cont'd)

Independence model

If X and Y are independent, the following conditions are satisfied for

$$\forall i = 1, \dots, I, \forall j = 1, \dots, J:$$

$$\pi_{ij} = \pi_{i+}\pi_{+j}, \quad \mu_{ij} = n\pi_{ij} = n\pi_{i+}\pi_{+j} \Rightarrow \log(\mu_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y$$

Expected frequencies: $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$

Log of expected frequency estimator: $\log(\hat{\mu}_{ij})$

Log-Linear Model for Contingency Table (Cont'd)

Example

For $I \times 2$ and contingency table, find estimators for odds and regression coefficient for $X = i$ when X and Y are independents

Probabilities

$$\pi_{ij} = \frac{\mu_{ij}}{\sum \sum \mu_{ij}}, \quad \mu_{ij} = \exp(\lambda + \lambda_i^X + \lambda_j^Y)$$

Odds

$$\log\left(\frac{P(Y=1)}{P(Y=0)}\right) = \log\frac{\mu_{i1}}{\mu_{i2}} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y$$

Odds is $\exp(\lambda_1^Y - \lambda_2^Y)$. If it does not include i , odds ratio becomes 1

Log-Linear Model for Contingency Table (Cont'd)

(Practice) Race by Belief in Life after Death data

	Yes	No
White	1339	300
Black	260	55
Others	88	22

```
> afterlife
  Race Belief Count
1 white  yes  1339
2 white  no   300
3 black  yes   260
4 black  no    55
5 others yes    88
6 others no    22
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.00032	0.10611	28.28	<2e-16	***
Racewhite	2.70136	0.09849	27.43	<2e-16	***
Raceblack	1.05209	0.11075	9.50	<2e-16	***
Beliefyes	1.49846	0.05697	26.30	<2e-16	***

```
require(glm2)
```

```
## Afterlife
```

```
afterlife <- data.frame(
```

```
  Race = c("white", "white", "black", "black", "others", "others"),
```

```
  Belief = rep(c("yes", "no"), 3),
```

```
  Count = c(1339, 300, 260, 55, 88, 22)
```

```
)
```

```
afterlife
```

```
afterlife$Race <- factor(afterlife$Race, levels=c("others", "white", "black"))
```

```
afterlife_model <- glm(Count~Race+Belief, data=afterlife, family="poisson")
```

```
summary(afterlife_model)
```

Interpretation

Odds for afterlife is $\exp(\lambda_1^Y - \lambda_2^Y) = \exp(1.50) = 4.48$

Log-Linear Model for Contingency Table (Cont'd)

Saturated log-linear model

Parameterizing interaction term

Dependence is indicated by interaction term

$$\log(\mu_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$$

$\{\lambda_{ij}^{XY}\}$: indicates the amount of dependence between X and Y

Number of parameters (0-1 / sum-to-1 restrictions)

Intercept (λ): 1, X (λ_i^X): $I - 1$, Y (λ_j^Y): $J - 1$, XY (λ_{ij}^{XY}): $(I - 1)(J - 1)$

Total number of parameters: $1 + (I - 1) + (J - 1) + (I - 1)(J - 1) = IJ$

Log-Linear Model for Contingency Table (Cont'd)

Saturated log-linear model

Hypothesis for interaction term

$$\text{Log-linear model } \log(\mu_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$$

Null hypothesis for testing independence: for any i, j , $\lambda_{ij}^{XY} = 0$

$(I - 1)(J - 1)$ parameters can be defined for interaction and likelihood ratio tests following chi-square distribution with $(I - 1)(J - 1)$ degrees of freedom can be used to test the independence.

Log odds ratio: 2×2 contingency table

$$\log\left(\frac{\mu_{11}\mu_{22}}{\mu_{12}\mu_{21}}\right) = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$$

Under null hypothesis, log-odds ratio is 0

Example

If log-odds ratio=0 for 2×2 contingency table, what are λ_{ij}^{XY} ?

Log-Linear Model for Contingency Table (Cont'd)

(Practice) Race by Belief in Life after Death data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.0910	0.2132	14.498	< 2e-16	***
Racewhite	2.6127	0.2209	11.829	< 2e-16	***
Raceblack	0.9163	0.2523	3.632	0.000281	***
Beliefyes	1.3863	0.2384	5.816	6.03e-09	***
Racewhite:Beliefyes	0.1096	0.2468	0.444	0.656946	
Raceblack:Beliefyes	0.1671	0.2808	0.595	0.551889	

```
afterlife_interaction_model <-  
glm(Count~Race*Belief,data=afterlife,family="poisson")  
summary(afterlife_interaction_model)
```

Interpretation

Estimates $\hat{\lambda} = 3.091$, $\hat{\lambda}_{white}^{race} = 2.613$, $\hat{\lambda}_{black}^{race} = 0.9163$, $\hat{\lambda}_{yes}^{belief} = 1.386$,
 $\hat{\lambda}_{white,yes}^{race \times belief} = 0.1096$, $\hat{\lambda}_{black,yes}^{race \times belief} = 0.1671$

Odds ratio for afterlife belief between white and others: $\exp(0.1096) = 1.12$

Odds ratio for afterlife belief between black and others: $\exp(0.1671) = 1.18$

Odds ratio for afterlife belief between white and black: $\exp(0.1096 - 0.1671) = 0.94$

Log-Linear Model of Three-Way Contingency Table

Log linear model

Saturated model

$$\text{Model: } \log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ} + \lambda_{ijk}^{XYZ}$$

Notation: (XYZ)

Homogeneous association model

$$\text{Model: } \log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$$

Notation: (XY, YZ, XZ)

Property

No three-way interaction

Conditional odd ratios between two variables are always same

$$\log \theta_{XY(k)} = \log \left(\frac{\mu_{11k} \mu_{22k}}{\mu_{12k} \mu_{21k}} \right) = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Log linear model

Conditional independence model

Model: assume X and Y are independent when Z is conditioned ($X \perp Y|Z$)

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ} \Leftrightarrow \pi_{ij|k} = \pi_{i+|k} \pi_{+j|k}$$

Notation: (YZ, XZ)

Properties

It has two interaction terms, and there is no interaction term between X and Y

It does not mean the marginal independence between X and Y . It can be marginally dependent

$$\text{cov}(X, Z) \neq 0, \text{cov}(Y, Z) \neq 0 \Rightarrow \text{cov}(X, Y) \neq 0$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Log linear model

Conditional independence model

Properties

Marginal and conditional ORs between X and Z are same (for Y and Z , they are same)

$$\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k} \Leftrightarrow \pi_{ijk} = \frac{\pi_{i+} \pi_{+j} \pi_{+k}}{\pi_{++k}}$$
$$\Rightarrow \theta_{XZ(j)} = \frac{\pi_{1j1}\pi_{2j2}}{\pi_{1j2}\pi_{1j2}} = \frac{(\pi_{1+1}\pi_{+j1}/\pi_{++1})(\pi_{2+2}\pi_{+j2}/\pi_{++2})}{(\pi_{2+1}\pi_{+j1}/\pi_{++1})(\pi_{1+2}\pi_{+j2}/\pi_{++2})} = \frac{\pi_{1+1}\pi_{2+2}}{\pi_{2+1}\pi_{1+2}} = \theta_{XZ}$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Log linear model

Joint independence model

Model: Y and (X, Z) are jointly independent ($Y \perp (X, Z)$)

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$$

Notation: (XZ, Y)

Property

There exists a interaction term between X and Y

No interaction term between X and Y

$\Leftrightarrow X$ and Y are conditionally independent

No interaction term between Y and Z

$\Leftrightarrow Y$ and Z are conditionally independent

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Log linear model

Mutual independence model

Model: $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$

Notation: (X, Y, Z)

Property

All variables are mutually independent

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Log linear model

Marginal independence model

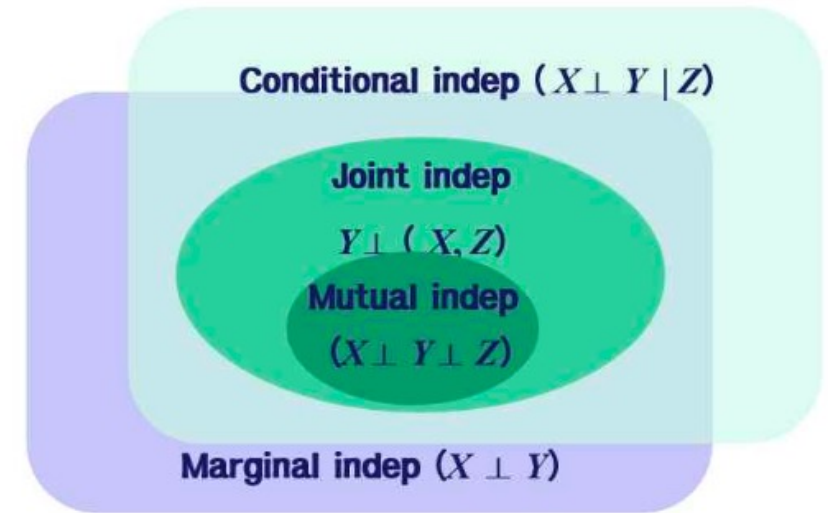
$$\pi_{ij+} = \pi_{i++}\pi_{+j+}$$

Property

Conditional independence between X and Y does not indicate the marginal independence

Marginal independence between X and Y does not indicate conditional independence

$Y \perp (X, Z)$ (joint independence) $\Rightarrow X \perp Y|Z$ and $Y \perp Z|X$



Log-Linear Model of Three-Way Contingency Table (Cont'd)

Example

Check whether each statement is true:

(1) X, Y, Z : mutual independence $\Rightarrow$ joint independence

(2) $Y \perp (X, Z)$ (joint independence) $\Rightarrow X \perp Y|Z$ and $Y \perp Z|X$

(3) $X \perp Y|Z$ and $Y \perp Z|X \Rightarrow X \perp Y$ and $Y \perp Z$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Example

(sol)

$$(1) \quad \pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k} \Rightarrow \pi_{ijk} = \pi_{i+k}\pi_{+j+}$$

$$(2) \quad \pi_{ij|k} = \frac{\pi_{ijk}}{\pi_{++k}} = \frac{\pi_{i+k}\pi_{+j+}}{\pi_{++k}} = \left(\frac{\pi_{i+k}}{\pi_{++k}} \right) \pi_{+j+} = \pi_{i+|k}\pi_{+j|k}$$

$$\because \pi_{+j|k} = \pi_{++k} \frac{\pi_{+j+}}{\pi_{++k}} = \pi_{+j+}$$

$$(3) \text{ By } \pi_{ijk} = \pi_{i+k}\pi_{+j+},$$

$$\pi_{ij+} = \pi_{i++}\pi_{+j+} \quad \text{and} \quad \pi_{+jk} = \pi_{++k}\pi_{+j+}$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Hierarchical model for three-way contingency table

Model	N of parameters	DF of G^2	Meaning
(XYZ)	IKJ	0	Saturated model
(XY, XZ, YZ)	$1 + (I - 1) + (J - 1)$ $+ (K - 1) + (I - 1)(J - 1)$ $+ (J - 1)(K - 1)$ $+ (I - 1)(K - 1)$	$(I - 1)(J - 1)(K - 1)$	
(XZ, YZ)	$1 + (I - 1) + (J - 1)$ $+ (K - 1) + (I - 1)(K - 1)$ $+ (J - 1)(K - 1)$	$K(I - 1)(J - 1)$	$X \perp Y Z$
(XZ, Y)	$1 + (I - 1) + (J - 1)$ $+ (K - 1) + (I - 1)(K - 1)$	$(K - 1)(IJ - 1)$	$Y \perp (X, Z)$ $Y \perp X Z$ $Y \perp Z X$ Marinal OR between $X - Z$ = conditional ORs
(X, Y, Z)	$1 + (I - 1) + (J - 1)$ $+ (K - 1)$	$IKJ - I - J - K + 2$	$X \perp Y \perp Z$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

(Practice) Marijuana data

Drinking (A)	Smoking (C)	Marijuana (M)	
		Yes	No
Yes	Yes	911	538
	No	44	456
No	Yes	3	43
	No	2	279

A	C	M	Observed	Log-linear model		
				(A,C,M)	(AM,CM)	(AC,AM,CM)
Y	Y	Y	911	539.98	909.24	910.38
		N	538	740.23	438.84	538.62
	N	Y	44	282.09	45.76	44.62
		N	456	386.70	555.16	455.38
N	Y	Y	3	90.60	4.76	3.62
		N	43	124.19	142.16	42.38
	N	Y	2	43.33	0.24	1.38
		N	279	64.88	179.84	279.62

```
## Marijuana
marijuana <- data.frame(
  Drinking = c(rep("yes",4),rep("no",4)),
  Smoking = rep(c(rep("yes",2), rep("no",2)),2),
  Marijuana = rep(c("yes","no"),4),
  Count = c(911,538,44,456,3,43,2,279)
)
marijuana

marijuana_m1 <- glm(Count~Drinking+Smoking+ Marijuana,data=
marijuana ,family="poisson")
marijuana_m1$fitted.values

marijuana_m2 <- glm(Count~Drinking*Marijuana
+Smoking*Marijuana,data= marijuana ,family="poisson")
marijuana_m2$fitted.values

marijuana_m3 <-
glm(Count~Drinking*Smoking+Drinking*Marijuana
+Smoking*Marijuana,data= marijuana ,family="poisson")
marijuana_m3$fitted.values
```

Log-Linear Model of Three-Way Contingency Table (Cont'd)

(Practie) Marijuana data

Interpretation

(1) (AM,CM)

When M is conditioned on yes or no, conditional ORs between A-C are

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} \approx 1.0 \approx \frac{438.84 \times 179.84}{555.16 \times 142.16}$$

Marginal ORs between A-C

$$\frac{(909.24 + 438.84)(0.24 + 179.84)}{(45.76 + 555.16)(4.76 + 142.16)} = 2.7$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

(Practie) Marijuana data

Interpretation

(2) (AC,AM,CM)

When M is conditioned on yes or no, condiaional ORs between A-C are

$$\frac{910.38 \times 1.38}{44.62 \times 3.62} = 7.78 \quad \frac{538.62 \times 279.62}{455.38 \times 42.38} = 7.80$$

Marginal ORs between A-C

$$\frac{(910.38 + 538.62)(1.38 + 279.62)}{(44.62 + 455.38)(3.62 + 42.38)} = 17.70$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Interpretation of parameters

Odds ratio

Marginal ORs

$$\theta_{ij}^{XY} = \frac{\mu_{i,j,+} \mu_{i+1,j+1,+}}{\mu_{i+1,j,+} \mu_{i,j+1,+}}$$

When $Z = k$, conditional ORs

$$\theta_{ij(k)}^{XY} = \frac{\mu_{i,j,k} \mu_{i+1,j+1,k}}{\mu_{i+1,j,k} \mu_{i,j+1,k}}$$

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Interpretation of parameters

2×2×2 contingency table

Relationship between log linear model parameter and ORs

$$\frac{1}{8} \log \frac{\theta_{11(1)}}{\theta_{11(2)}} = \lambda_{111}^{XYZ}$$

⇒ By the parameter restrictions,

$$\lambda_{11}^{XY} = -\lambda_{12}^{XY} = -\lambda_{21}^{XY} = \lambda_{22}^{XY}, \quad \lambda_{111}^{XYZ} = -\lambda_{121}^{XY} = -\lambda_{211}^{XY} = \lambda_{221}^{XY}$$

Thus, (1) $\log \theta_{11(1)} = \log \mu_{11k} + \log \mu_{22k} - \log \mu_{21+} - \log \mu_{12k}$ (2) $\log \theta_{11(2)} = 4\lambda_{11}^{XY} - 4\lambda_{111}^{XYZ}$

$$\begin{aligned} &= (\lambda + \lambda_1^X + \lambda_1^Y + \lambda_1^Z + \lambda_{11}^{XY} + \lambda_{11}^{YZ} + \lambda_{11}^{XZ} + \lambda_{11}^{XYZ}) \\ &+ \lambda + \lambda_2^X + \lambda_2^Y + \lambda_1^Z + \lambda_{22}^{XY} + \lambda_{21}^{YZ} + \lambda_{21}^{XZ} + \lambda_{221}^{XYZ} \\ &- \lambda - \lambda_2^X - \lambda_1^Y - \lambda_1^Z - \lambda_{21}^{XY} - \lambda_{11}^{YZ} - \lambda_{21}^{XZ} - \lambda_{211}^{XYZ} \\ &- \lambda - \lambda_1^X - \lambda_2^Y - \lambda_1^Z - \lambda_{12}^{XY} - \lambda_{21}^{YZ} - \lambda_{21}^{XZ} - \lambda_{121}^{XYZ} \\ &= 4\lambda_{11}^{XY} + 4\lambda_{111}^{XYZ} \end{aligned}$$

Parameters for log linear model can be always interpreted by ORs

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Example

Prove that if $\theta_{11(1)} = \dots = \theta_{11(K)}$, $\frac{1}{4} \log \theta_{11(k)} = \lambda_{11}^{XY}$

High-way contingency table

It means there are three or more variables

For high-way contingency table, log linear model is useful for its interpretation and model building

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Collapsibility

Independence graph

Independence graph consists of vertex and edge

- (1) Vertex: each vertex indicates a variable and their number is equal to the number of variables
- (2) Edge: If there is an edge between two vertex, two variables are not independent, and otherwise they are independent
- (3) Path: If two vertices can be connected by edge, those edges are called path between both vertices
- (4) Separated: X and Y is separated by $Z \iff$ All paths between X and Y include Z

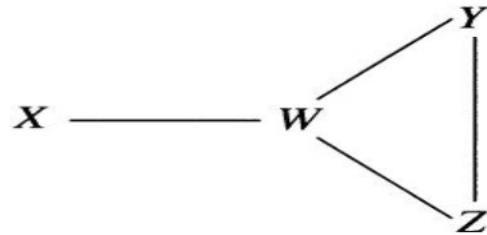
Log-Linear Model of Three-Way Contingency Table (Cont'd)

Collapsibility

Example

State the relationship among W, X, Y, Z for the following independence graphs

①



- ① (WX, WY, WZ, YZ) ,
Path between X and Z : (X, W, Z) , (X, W, Y, Z)
 X and $\{Y, Z\}$ can be separated by W .

②



- ② (WX, XY, YZ)
Path between X and Z : (X, Y, Z)
 X and Z can be separated by Y .

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Collapsibility

Marginal and conditional ORs are same

Requirement for collapsibility

Association is not changed when partial tables are pooled

Example

(1) If $X \perp Y|Z$ for three-way contingency table, is it collapsible?

$$\because X \perp Y|Z \Rightarrow \theta_{XZ(j)} = \theta_{XZ}, \theta_{YZ(i)} = \theta_{YZ}$$

Thus $X - Z$ and $Y - Z$ are collapsible, however, $X - Y$ is not collapsible

(2) The homogeneous association, (XY, XZ, YZ) , is collapsible?

Log-Linear Model of Three-Way Contingency Table (Cont'd)

Collapsibility and independence graph

Independence graph for $X \perp Z|Y$

$$X \text{ — } Y \text{ — } Z$$

$X - Y$ and $Y - Z$ are collapsible

Goodness of fit test

Statistics for goodness of fit

Pearson chi-square test X^2

Likelihood ratio test G^2

Example

The following is the result from Marihuana data. A: drinking, C: smoking, M: marihuana. Choose the best model.

· Data description

Drinking (A)	Smoking (C)	Marihuana (M)	
		Yes	No
Yes	Yes	911	538
	No	44	456
No	Yes	3	43
	No	2	279

· Result

Model	df	X^2	G^2	P
(A,C,M)	4	1441.39	1286.00	0.00
(AM,CM)	2	177.61	187.75	0.00
(AC,AM,CM)	1	0.401	0.37	0.54
(ACM)	0	0	0	

(AC,AM,CM) is the best

Goodness of fit test (Cont'd)

Residual

Pearson residual

Adjusted Pearson residual

If assumptions are satisfied, it follows the standard normal distribution

Cells with absolute adjusted residual > 2 or 3 indicates the poor fit of the model

Goodness of fit test (Cont'd)

(Practice) Marihauna data

A	C	M	Adjusted residual	
			(AM,CM)	(AC,AM,CM)
Y	Y	Y	3.7	0.6
		N	12.8	-0.6
	N	Y	-3.7	-0.6
		N	-12.8	0.6
N	Y	Y	-3.7	-0.6
		N	-12.8	0.6
	N	Y	3.7	0.6
		N	12.8	-0.6

```
rstandard (marijuana_m2)
rstandard (marijuana_m3)
```


Goodness of fit test (Cont'd)

(Practice) Marihauna data

Interpretation

(1) (AM,CM): it assumes that A-C are independent

$$X^2 = 177.61 = 3.7^2 + 12.8^2$$

For subjects who did not experienced marihauna, the absolute value of adjusted standardized residuals is 12.8 and much larger than 3.7 for subjects who experienced it

Positive residuals are observed for subjects who experienced both drinking and smoking, or neither

Positive residuals indicates the observed frequency is larger than frequencies estimated under the assumption of independence between smoking and drinking

(2) (AC,AM,CM)

$$X^2 = 0.401 = 0.63^2$$

Every cells have small adjusted standardized residuals, and the model appropriately fit data

Goodness of fit test (Cont'd)

(Practice) Marihauna data

Interpretation

(2) (AC,AM,CM)

$$X^2 = 0.401 = 0.63^2$$

Every cells have small adjusted standardized residuals, and the model appropriately fit data

Goodness of fit test (Cont'd)

Partial association

Model selection: likelihood ratio test

M_1 : reduced model, M_2 : full model ($M_1 \subset M_2$)

$$\Rightarrow G^2(M_1|M_2) = G^2(M_1) - G^2(M_2) \sim \chi^2(df)$$

df = N of parameters for M_2 - N of parameters for M_1

Choose the better model among (AM,CM) and (AC,AM,CM)

Model	df	X^2	G^2	P-value
(A,C,M)	4	1441.39	1286.00	0.00
(AM,CM)	2	177.61	187.75	0.00
(AC,AM,CM)	1	0.401	0.37	0.54
(ACM)	0	0	0	

Goodness of fit test (Cont'd)

Partial association

Interpretation

$$H_0 : \lambda^{AC} = 0 \text{ vs } H_1 : \text{not } H_0$$

$$\begin{aligned} G^2[(AM, CM)|(AC, AM, CM)] &= G^2(AM, CM) - G^2(AC, AM, CM) \\ &= 187.75 - 0.38 = 187.38 \sim \chi^2(df = 2 - 1) \end{aligned}$$

Thus, null hypothesis $H_0 : \lambda^{AC} = 0$ is rejected

Relationship between Log Linear Model and Logistic Model

Difference between two models

Logistic model: binary response variables and multiple explanatory variables

Logistic linear model: both response and explanatory variables are categorical variables

As long as variables are binary, then logistic regression can be utilized

Relationship between two models

Log linear model	Logistic model
(Y, XZ)	α
(XY, XZ)	$\alpha + \beta_i^X$
(YZ, XZ)	$\alpha + \beta_k^Z$
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$

Relationship between Log Linear Model and Logistic Model (Cont'd)

Relationship between two models

For instance, homogeneous association can be expressed by

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ij}^{XZ} + \lambda_{jk}^{YZ}$$

$$\Rightarrow \log \frac{\mu_{i1k}}{\mu_{i2k}} = (\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ})$$

$$\Rightarrow \log \frac{\pi_{i1k}}{\pi_{i2k}} = \alpha + \beta_i^X + \beta_k^Z \quad (\text{additive})$$